

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – Analytical Problem Solving Task

Course Code:	CSA0620	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving Task
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO2 Statement:	Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)		
Student Name:	Kamalesh T	Reg. No.:	192424228
Section / Batch:	2024	Date:	31/07/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Analytical Problem Solving Task	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1	100
GRAND TOTAL:			100 Marks

SECTION A – APPLICATION BASED QUESTIONS

Q1. Sorting Techniques – Selection Sort vs Bubble Sort

Analyze the performance of Selection Sort and Bubble Sort on the dataset [28, 16, 35, 11, 19].

Clearly interpret the problem and explain the working principles of both algorithms.

Apply both methods step-by-step and show intermediate results after each pass. Count the total number of comparisons and swaps in each case. Analyze their time complexities and compare their efficiency.

Interpret the results and justify which method is preferable for this dataset.

Code of Conduct

I P. Kamalesh T(192424228) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Kamalesh T



RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (100 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Analytical Problem Solving Task

Sorting Techniques - Selecting Sort vs Bubble Sort

Aim:

To analyze and compare the performance of Selection Sort and Bubble Sort on the dataset [28, 16, 35, 11, 19]

Problem Interpretation:

Sorting is the process of arranging data in ascending or descending order. Here, the objective is to sort the given list in ascending order using two different Sorting algorithms and compare their performance based on:

- Step by step execution
- No of Comparisons
- No of Swaps
- Time Complexity
- Overall efficiency

1. Selection Sort

Working principle

Selection Sort repeatedly finds the smallest element from the unsorted portion of the array & places it at the beginning.

Step-by-Step Execution

Initial Array

[28, 16, 35, 11, 19]

Pass 1

Smallest element = 11

Swap 28 & 11

[11, 16, 35, 28, 19]

Pass 2

[16, 35, 28, 19]

Smallest = 16

Already in correct position

Pass 3

Remaining array:

[28, 35]

Smallest = 28

No swap required

[11, 16, 19, 28, 35]

Pass 4

Remaining array:

[35, 28, 19]

Smallest = 19

swap 35 & 19

[11, 16, 19, 28, 35]

Selection Sort Table

Pass	Array After Pass
Initial	28, 16, 35, 11, 19
1	11, 16, 35, 28, 19
2	11, 16, 35, 28, 19
3	11, 16, 19, 28, 35
4	11, 16, 19, 28, 35

Comparisons and Swaps

Comparisons

$$\rightarrow \text{Pass 1} = 4$$

$$\rightarrow \text{Pass 2} = 3$$

$$\rightarrow \text{Pass 3} = 2$$

$$\rightarrow \text{Pass 4} = 1$$

Swaps

$$\rightarrow \text{Pass 1} = 1$$

$$\rightarrow \text{Pass 2} = 0$$

$$\rightarrow \text{Pass 3} = 1$$

$$\rightarrow \text{Pass 4} = 0$$

$$\text{Total Swaps} = 2$$

2. Bubble Sort

Working Principle

Bubble Sort repeatedly compares adjacent element and swaps them if they are in the wrong order. After every pass, the largest element moves to the end.

Step by step Execution

Initial Array

[28, 16, 35, 11, 19]

Pass 1

28 > 16 → swap

[16, 28, 35, 11, 19]

35 > 11 → swap

35 > 19 → swap

[16, 28, 11, 19, 35]

Pass 2

28 > 11 → swap

[16, 11, 28, 19, 35]

28 > 19 → swap

[16, 11, 19, 28, 35]

Pass 3

16 > 11 → swap

[11, 16, 19, 28, 35]

Pass 4

No swaps

[11, 16, 19, 28, 35]

Bubble Sort Table

Pass	Array After Pass
Initial	
1	28, 16 35 11 19
2	16 28 11 19 35
3	16 11 19 28 35
4	11 16 19 28 35
	11 16 19 28 35

Comparisons and Swaps

Comparisons

- Pass 1 = 4
- Pass 2 = 3
- Pass 3 = 2
- Pass 4 = 1

→ Total Comparisons = 10

Swaps

- Pass 1 = 3
- Pass 2 = 2
- Pass 3 = 1
- Pass 4 = 0

→ Total Swaps = 6

Time Complexity Analysis

Algorithm	Best case	Average case	Worst case	Space complexity
Selection Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$	$O(1)$
Bubble Sort	$O(n)$	$O(n^2)$	$O(n^2)$	$O(1)$

Result Interpretation:

- Both algorithms required 10 comparisons for the given dataset
- Selection Sort performed only 2 swaps, while Bubble Sort performed 6 swaps

Conclusion:

For the dataset [28, 16, 35, 11, 19], Selection Sort is preferable because it achieves the same sorted output with significantly fewer swaps (2 compared to 6), while both algorithms perform the same number of comparisons (10).